

1. Check the correct answer of the follow: [5 Marks]

- a. Process Control Block
 - i. includes information on the process's state
 - ii. is stored in the same address space of the process
 - iii. determines which process is to be executed next
 - iv. is an example of a process queue
- b. The list of processes waiting for a particular I/O device is called a(n)
 - i. ready queue
 - ii. standby queue
 - iii. interrupt queue
 - iv. device queue
- c. is the number of processes that are completed per time unit.
 - i. Turnaround time
 - ii. Response time
 - iii. CPU utilization
 - iv. Throughput
- d. scheduling is approximated by predicting the next CPU burst with an exponential average of the measured lengths of previous CPU bursts
 - i. RR
 - ii. SJF
 - iii. FCFS
 - iv. Multilevel queues
- e. Which of the following scheduling algorithms must be non preemptive?
 - i. RR
 - ii. SJF
 - iii. FCFS
 - iv. Priority Algorithms.
- f. Which processes scheduling algorithm gives the minimum average waiting time?
 - i. Priority scheduling.
 - ii. FCFS.
 - iii. SJF.
 - iv. None of the above.
- g. Operating system is a program that:
 - i. Makes the computer more convenient and more efficient and manage all the computer resources.
 - ii. Converts the high level language into machine language.
 - iii. Manages database.
 - iv. All the above.

- h. A small CPU time quantum for process scheduling is:
 - i. To increase the CPU performance.
 - ii. To increase the response time.
 - iii. To increase throughput.
 - iv. None of the above.
- i. Some targets of the processes scheduler are
 - i. Decrease throughput.
 - ii. Decrease response time.
 - iii. Increase turnaround time
 - iv. All the above
- j. A trap is a (n)..... interrupt
 - i. event handler
 - ii. Interrupt handler
 - iii. software-generated
 - iv. hardware-generated

2. [5 Marks]

- a. Describe the actions taken by a kernel to context-switch between processes.
- b. What is the motivation to operate computer system in dual modes, then state when mode switching occurs?
- c. Which of the following instruction working only in the privileged mode? Then give a reason explaining the motivation for your answer.
(Check all the correct answers)
 - i. Instruction which sets the quantum.
 - ii. Instruction to multiply two numbers.
 - iii. Read the clock.
 - iv. Turn off interrupt.
- d. What is the difference between trap and interrupt and for which purpose each one is used?

3.

- a. Give the process state model and determine the different events on which transition from state to another take place? [2 Marks]

- b. Consider the following set of processes, with the length of CPU burst time given in milliseconds: [3 Marks]

- i. Draw the Gantt charts illustrating the execution of these processes using preemptive SJF and nonpreemptive priority scheduling. Calculate response, turnaround and waiting times for each process and the average time for each in the tables down here.

Process	arrival	Burst	Priority
P ₁	0	10	2
P ₂	3	8	3
P ₃	4	4	1
P ₄	5	4	4

Big number for priority means high priority

Draw Gantt Chart for each of the scheduling Algorithm above its corresponding table

Process	response time	TurnAround time	Waiting time
P ₁			
P ₂			
P ₃			
P ₄			
Average			

Preemptive SJF

Process	response time	TurnAround time	Waiting time
P ₁			
P ₂			
P ₃			
P ₄			
Average			

Nonpreemptive Priority